

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	28 June 2026
Team ID	SWTID-2026-7871
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import and Integration	Import agricultural dataset (1997-2021) into Tableau Validate data structure and attributes Establish relationships among dataset fields.
FR-2	Data Analysis and Processing	Calculate Total Production, Yield and Cultivated Area Create calculated fields and aggregations Perform year-wise and state-wise analysis
FR-3	Dashboard Development	Create agricultural Overview Dashboard Create state wise production dashboard Perform year wise and state wise analysis
FR-4	Data Visualization	Develop Bar Charts, Line charts, Maps and KPI Cards Design comparative analysis charts
FR-5	Interactive Features	Implement filters for State, Year, Crop and Season Enable dashboard navigation
FR-6	Story Development	Create Tableau Story with key insights Present recommendations and conclusions
FR-7	Data Publishing	Publish Dashboards on Tableau Public Generate shareable dashboard links
FR-8	Reporting and Insights	Generate production trend reports Provide insights for farmers, researchers and policymakers

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboards should provide an intuitive and user-friendly interface with interactive filters, charts, and visualizations that can be easily understood by farmers, researchers, and policymakers.
NFR-2	Security	The project uses publicly available agricultural data and is published through Tableau Public, ensuring safe access without requiring user

		authentication or sensitive data handling
NFR-3	Reliability	The system should consistently display accurate agricultural insights and maintain data integrity across all dashboards and story visualizations.
NFR-4	Performance	The dashboards should load efficiently and respond quickly to filter selections, drill-down operations, and visualization rendering
NFR-5	Availability	The dashboards and stories published on Tableau Public should remain accessible to users anytime through an internet connection
NFR-6	Scalability	The solution should support the addition of future agricultural datasets, new visualizations, and extended analysis without major redesign